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- Figure 1. The single-cycle datapath with the control unit

PC's Clk.- to-Q	I-Mem.	Add	Mux	ALU	Ctrl. Unit	ALU- Ctrl.	Regs. Access	Data- Mem.	Sign- extend	AND- Gate	Shift- left-2
0ps	200ps	80ps	60ps	150ps	250ps	40ps	120ps	180ps	30ps	50ps	10ps

Instr.	Opcode	Funct
add	0 ₁₀	32 ₁₀
sub	0 ₁₀	34 ₁₀
beq	4 ₁₀	
lw	35 ₁₀	
sw	43 ₁₀	

- Ans: 770 ps

1.2. (5%) What is the **clock cycle time** of this processor that support the instructions listed in **Table 2**?

無腦沖 $t_w = 900\text{ps}$.

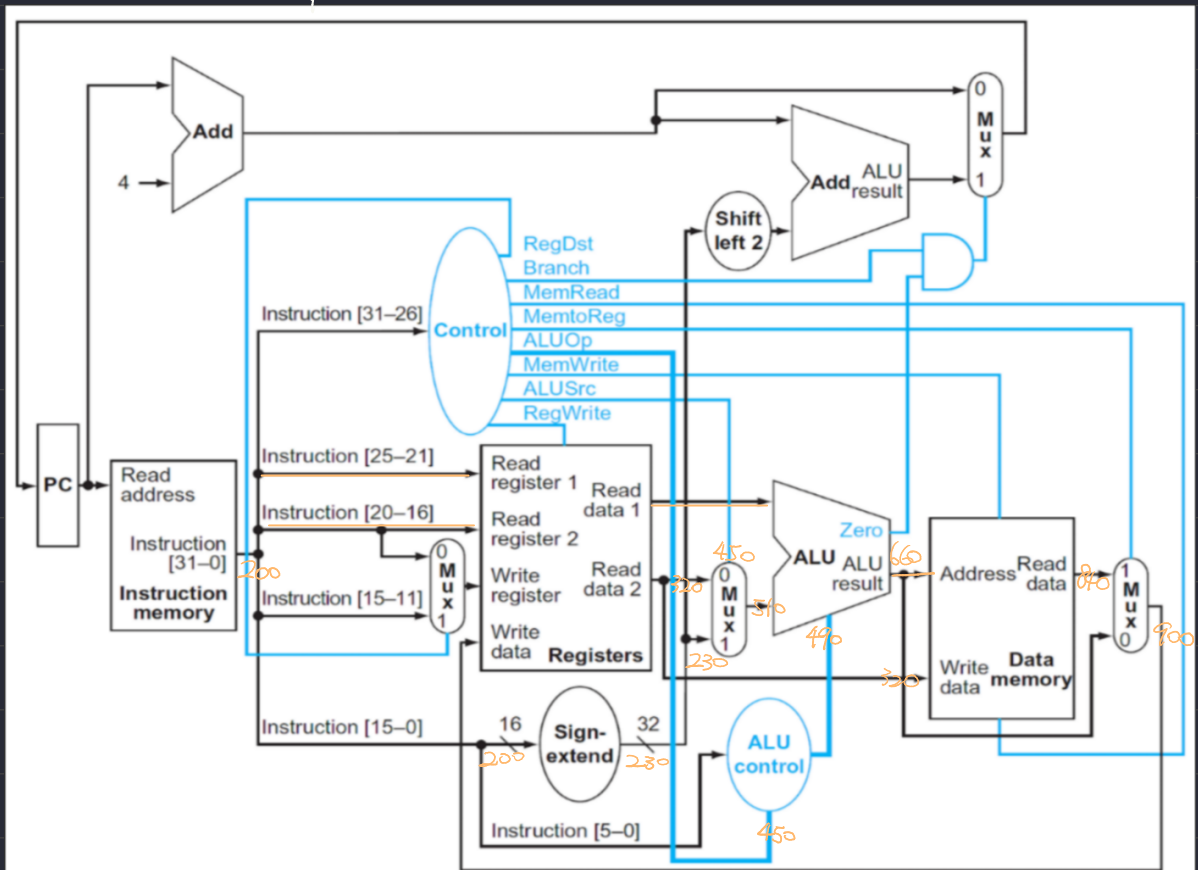


Table 1. Latencies of the functional units on the datapath

PC's Clk.-to-Q	I-Mem.	Add	Mux	ALU	Ctrl. Unit	ALU-Ctrl.	Regs. Access	Data-Mem.	Sign-extend	AND-Gate	Shift-left-2
0ps	200ps	80ps	60ps	150ps	250ps	40ps	120ps	180ps	30ps	50ps	10ps

2. (5%) Assume -33 and 95 are signed 8-bit decimal integers stored in two's complement format. Calculate -33-95 Is there overflow, underflow or neither ? (3%) Explain why? (2%)

$$-33 - 95 = -128$$

$$\begin{aligned} 8\text{bit } 2^{\text{complement}} &= -2^7 \sim 2^7-1 \\ &= -128 \sim 127 \end{aligned}$$

A: neither

3. (10%) Please convert (1) the decimal number **0.84375** to IEEE single-precision floating point format, (2) the binary number **110.11011** to decimal format.

$$(1) 0.84375 = 0.11011_{(2)}$$

$$= 1.1011 \times 2^{-1}$$

$$\text{Ans: } \begin{array}{c} 0 \ 0111110 \quad 1011 \ 0000 \ 0000 \ 0000 \ 0000 \ 0000 \\ \hline \text{S} \quad \text{EXPONENT (12s)} \quad \text{Fraction (23bits)} \end{array}$$

$$\begin{array}{r} 0.84375 \\ \times \quad 2 \\ \hline 1.68750 \dots 1 \\ \times \quad 2 \\ \hline 3.37500 \dots 1 \\ \times \quad 2 \\ \hline 6.75000 \dots 0 \\ \times \quad 2 \\ \hline 13.50000 \dots 1 \\ \times \quad 2 \\ \hline 27.00000 \dots 1 \end{array}$$

(2)

$$\begin{aligned} &6 + \frac{1}{2} + \frac{1}{4} + \frac{1}{16} + \frac{1}{32} \\ &= 6 + \frac{16+8+2+1}{32} \\ &= 6 + \frac{27}{32} \\ &= 6.84375 \end{aligned}$$

4. (10%) Describe how Carryout are generated step by step to determine the last Carryout (CarryOut4) in the following example (please use the concept of carry-lookahead)

x: 1101

y: 0110

p: 1111

g: 0100

$$C4 = g3 + (p3 \cdot g2) + (p3 \cdot p2 \cdot g1) + (p3 \cdot p2 \cdot p1 \cdot g0) + (p3 \cdot p2 \cdot p1 \cdot p0 \cdot c0) = 1$$

skip

5. (10%) Sketch a 4-bits Adder-Subtractor-NAND-Overflow_detection Circuit using 4 Full-Adder (FA) and 5 XOR gates. The input are (B.B.B.B.), (A.A.A.A.) and the signal ctrl controls the add/sub operation (ctrl = 00, 01, 10 for **NAND/** Subtractor /Adder). In your design try to use minimum number of the following basic logic gates (1-bit adders, XOR, NAND, MUX). (Draw the design of 1bit ALU with detail will get (5%), then use the block of design to draw the full circuit will get remained (5%).

Ans:

skip

6. (10%) Please calculate the product of the decimal **-7 x 5** by using Booth's Algorithm. (check 2 bit : 00 (0) , 01(1), 10(-1), 11(0))

Ans:

$$-7 \text{ (Multiplicand)} = 1111 \ 1001$$

$$5 \text{ (Multiplier)} = 0000 \ 0101 \Rightarrow 00001(-1)1(-1)$$

$$-35 \text{ (Product)} = 1101 \ 1101$$

11111001
x)00001-11-1
00000111
11111001
00000111
11111001
100111011101(-35)

skip

7. (10%) When performing arithmetic addition and subtraction, overflow might occur. Fill in the blanks in the following table of overflow conditions for addition and subtraction.

Operation	Operand A	Operand B	Result indicating overflow
$A + B$	≥ 0	≥ 0	(a) ____ < 0 ____ or -
$A + B$	< 0	< 0	(b) ____ ≥ 0 ____ or +
$A - B$	≥ 0	< 0	(c) ____ < 0 ____ or -
$A - B$	< 0	≥ 0	(d) ____ ≥ 0 ____ or +